

Social constraints (Lecture) ~ Avel Guénin-Carlut ~

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hello and welcome everyone it's October 18th 2023 we're in the course active inference for the social sciences and today we're in the lecture by the course co-organizer and that will kick off the final leg of the course which is AAL gwiin Caro's lecture social constraints and the existential what makes us human so we'll be looking forward to a lecture followed by a discussion so if you have any questions please feel free to add them in the live chat or submit to the course syllabus and also at the end we'll talk about how we're going to conclude this course so with that thank you to you evil Hello uh sir do you see the slides correctly everything is fine I will take sence as a yes uh so uh the the role of this session is basically to uh to show you the inspiring intuition behind the all course and to show very concretely well very abstractly but in my mind concretely why the specific formalism of active inference and social constraints is relevant to understand the human species and the stult evolution of you know things that we do so we'll talk of uh social constraints of the exist of the existential stance and of what makes us human so disclaimer uh this paper this presentation is closely aligned with upcoming paper which whose name is like the same I think is tentative for now and whose cter whose man on job is a c so uh first let us ask basic questions about the topic at hand what is is culture uh by psychologist sesa uh culture is typically used in one of three way culture can be information or behavior required socially so like if I am an ant and I figure out that you need to go somewhere because of feromon this is culture it can be social equate behavior that is shared within a group uh so for example if I am a nape and I figure out how to open nuts with a stone thing and the group just everyone does it the same way this is culture and it can be the Adaptive evolution of socially acquired behavior which is uh like fam well we have evidence of of it in the in in the wild for corvids and humans only so um the question is we know that culture is the human specific like it's it it is the case simply but depending on what we mean by culture it can entail it can describe behavior that is much very much

shared um with other species so the question a question that need to be answered is how specific are uh the human cultural mechanisms so um One Core thing to understand is that there is no such thing as the cultural mechanism the mechanisms of learning in humans they are uh culturally transmitted whether you will listen to I don't know Elder uh rapper um scientist depends on the cultural norms in which you exists and the specific way in which uh teacher student interaction are supposed to go is again culturally embedded and the type of information that people will try to figure out is or transmit is itself Cur embedded so we can just write the mechanism for human social learning this is like empirically there is known and conceptually I would don't I don't think the notion is coherent to begin with um so one core argument for this is We have basically um explicit linguistic rules about we who we should trust like listen to your parents and those are linguistic and language varies from culture to culture so this changes from culture to culture pretty straightforward and in general the uh practice of cultural submission of intentional teaching they uh they are varying between cultures and I do not want to get into it because I don't think I would say something that is accurate without notes and um we can however take a look at take a look at a behavior that is I think nearly Universal which is storytelling when people uh tell story they use stories to transmit norms and uh like in the specific uh culture of the a or AKA I I have misspelled it at least once I think it is agda like you have a like core role of Storytelling in telling people what to do and you have specific uh tales that were studied and that were associated to specific norms and these are very important because like here we have laws that are written somewhere and that like people in blue will come and hit you if you don't follow them but this is not something that is reveral so in a context where you do not have States you typically have no really other way to transmit Norms than you know talk about it uh and so from there everything is citt like the way the Norms the kind of norms can be transmitted given this way uh the way specific cues will be understood given specific Niche ET Etc so yes all this to say there is no Big C culture there is cultural transmission there is cultural Evolution but there is no specific mechanism or um you know natural type of culture in humans but uh there are uh categories of behavior that are very obviously human specifics and that are very obviously cultural as in direct contribution to culture for example fire it's obviously majorly important to culture because if you have fire you get a lot of calories if you get a lot of calories you get a big brain like can four brain but fire you know doesn't produce culture that's not how anything Works uh things that do work directly PR culture are language and Ro taking so language I think you are familiar with it you are using it now on the receiving side it is a practice that

basically embeds communicative intents into a specific set of symbols that can be letters that can be words like or it would be phon I guess I don't know the Phon or the word is a symbol but you have symbols and I Amed meaning and people use it and what is critical is that each of these symbol has a public meaning has a public like semantics and pragmatics if I tell you chair maybe we don't know precisely uh like what specific thing enters in the chair category maybe perceptual insation of chair will vary but like we are we agree on the basics of the chair we agree on what we can do with chairs we agree with a lot of thing implicitly by the use of this word and we can use the of syntax to manipulate this unit of meaning in a specific way I sit on a chair I'm talking an action that is relative to a specific object that is a chair and you understand what I mean if I was trying to you know mic it maybe I get it through maybe you would think I'm saying I'm crotching over uh bad what why not and so the language and the fact that it has a publically instanciated meaning is critical in you know keeping a coordination in the specific thing that people do and why they do it which is unambiguously cultural and you have another more subtle mechanism uh process I don't know phenomenon which is uh shared intentionality so by the work of mainly tomasel we know that uh chimps are not very social and they can learn they are but as intelligent as humans but at least young humans but they usually are not really interested in what other CHS are doing and most importantly they are not interested in um like sharing goals trying to figure out what is the end result that some other chimp is upholding and trying to like mesh in this plan to help him while in human from a very young age we are very exem motivated and we are very motivated to like figure out what we whatever we is are collectively doing and figure out what one's role is in that and do that and this is likely not you know a bona cognitive function that stands on its own but it's a behavior that is a cognitive process that is anemic in humans very rare if existing everywhere else and that is core to most theory of the of culture so these two things we know are human specifics up to maybe very small important details and we know are critically embedded in what we call culture and so what I will do uh from now on is to provide a specific theory of culture so the active inference of cultural landscapes that in my opinion basically explains or at least um meshes well with why those two specific phenomenons are so important from the apping of culture but it also means that culture is not information pery it's something that is more organic and that evolves more like you know biological system themselves evolve than population of gen and from these accounts we will go to um figure out what it means about Collective organization uh figure out what why we have gods and taxes and from there we will go to a more physical and pcal consideration and ask how basically this Theory can be written

within physics uh through the active inference formalism so let us go uh first I'm just presenting the theory of the active inference of clear landscape and I presenting a core concept the concept of C constraints that there from at least my treatment of that so active INF frence uh probably a lot of you know what this is but this goes on YouTube so I will re explain active inference is a theory of cognition where basically agents enact expectation of the world they have a world model and they make predictions over the world model and they strive for those predictions to remain coherent with what they actually observe and that means that the agent will you know update beliefs that are false if I uh hear moow and I think there is no Cuts I will then think that it is more likely that and on the other hand I will try to provoke to like realize uh prediction that I can of need so for example if I uh really want to hear Mo if I really expect to hear moo and I expect this to be like a basic steple of my life I will maybe figure out how to get a cats that works not well with cats but with drinking or eating yeah expect to be a thing that has water in it because else I die so when I'm thirsty I am Thirst by drinking and you have this basically duality of productive like normal what we call typically perception just trying to figure out what there is outside and updating our belief when we don't so I said perception but is more in inference and there is active inference which is uh like the realization of this inference through action and a core argument of active inference is that those two are uh same process they are the same process that is entailed by an information geometry so basically a space of belief over which the Dynamics of an agent uh unfold and the core feature of the this approach is uh the dissolution of agency as we typically understand it we typically understand agency to be like hey I'm an agent I want to do this and I will reason about what I want and I will reason about how to get it and will do the thing that I reasoned about reason is important it's not the basic cognitive function that analyze everything we do I do not think the statement is controversial IAL and in active inference the model we have is that like at multiple scale in multiple context we like again produce prediction over our sensor flow and then strive to realize those prediction so what that means is that direction are not constrained by will uh in the Jal case they're constrained by what looks more like a self model a model of on self and what on self would typically do in text such as as the one I'm exposed to um related concept is the notion of skill and intentionality so this is this uh skill intentionality thing is a theory that stands at the interface between active inference and ecological psychology which I will absolutely not get into and the core argument of skill intentionality is that when you are a skilled agent so agents that you know understand how to do things in the world you do not experience the world as a like uh represented space in 3D with uh 3D models in it and each

object is well defined and a lot of has a lot of properties that are embedded in a world model that like you have in a video game uh what you see is a landscape of basic opportunities for I will see a computer I know I can mash the keys keys I think you say keys and I know I can go to the internet and I know I can do this or that I see a door I know I can go through it if it's open I can close it I can go through it if it's closed and I can open it this is the basic opportunities that the door present and um we don't say a logical consequence but uh very coherent uh entailment active inference is that I will see minimal model of things around me that are based on the opportunities for Action that they present I will Orient my attention minimally to grasp what I can do with things and what the field perceptive field that I live in is a landscape of those affordances so the um picture is an extract from Fallout New Vegas which is the gr RPG please play it this is important only thing you need to keep in mind on this course and that shows you like a radio in like 3DF of radio and you have an action button that tells you whether you want to activate the radio and you have HPS and APS I have no idea what AP is and this looks more like a naive field than what you get into um really realistic presentation of like a a picture photo picture of your environment because you have the perception of the uh object like you have a TOD array that presents you know your vision of an object but it's embedded in things that tell you where you're looking at things that tell you what you can do with things things that tell you what are your States what are your basic state when you're engaging with this uh object and this is uh I think pretty nice uh illustr of what we could mean by perception and the landscape used you need LS sorry a could you repeat the last 30 seconds there it was just like it hung up okay just just repeat the last 30 seconds thank you so this is a picture from Fallout New Vegas which is a good game and I claim that it is a pretty good illustration of what we talk when we talk about ACC related perception because you see the object and with slash through the object you also see what opportunities for Action it presents in this case activate the radio and you see it in the way is embedded with your pH physiological States so the HP and AP and a core feature of that is that uh when I experience affordances I feel directly through the perception whether the affordance is good or bad uh if I see I don't know um a cliff and I I can fall off the cliff and die uh this will cause Fair the perception of that possibility of falling and dying will be negatively loaded if I see I don't know a hot bed in Winter and I know I can now rest after a day of work where I was cold I will be happy about it I will perceive the possibility in a like positively balanced way and so this is the case for like basic phys physiological thing but this is also the case for cultural norms and values we come to experience the world as a world of affordances and and this is

the same for the social world I will feel bad or good depending on what possibility for Action I will experience in a given context and um this is a picture of forest path near archical site that is um that is studied by the research group that funds me escape and like from a pure direct perception kind of way there is like uh very complex statistical feature that are made in the picture but if you are a human a thing that walks what you will perceive like feel directly experience directly is just a possibility to walk along the path and follow it and this is something that you feel is a property of the path I think you see a path you think I can go through a path you don't think there is a 3D space and in this 3D space there is a set of statistical feature one of which being a flat thing in the middle and among the enter space of action that they have one of them is to walk through the path like just do this and this and this and this you just feel like you can go through there and uh the way humans uh learn about each other uh the way humans citate and uh come to expect things about each other uh they can develop basically a sense of what is cultural possible or not possible in a given context and this uh sensation this perception of cultural possibilities it is embedded in the way we pay attention to things we will pay attention to different things according to like you know what is culturally rent in a given context in a given situation and this is a core argument of the um thinking through the minds account of cultural Evolution which was uh spearheaded by Maxwell rad and um this year forgot uh his first name and because of everything that we saw before because of the dissolved agency that we have under active in France the patterns cultural patterns in the FL of attention they will not simply be experienced as you know like abstract norms and value that one could or could not respect they will effectively constrain the thing you can pay attention to the outcome you expect in a way that will um Drive Your Action toward specific outcomes and so this cultural landscape of affordances they embed intivity as in the patterns that are relevant to explain normativity and intentionality they are in the landscape they are not like only at least in your brain cells and a core argument I would say is that uh the engagement with a cultural landscape it is D to existence of social constraints where cir constraints are understood as statistical patterns that will like causally act over what you do or cannot do so when I talk about the Str landscape of fores I am implicitly talking about a set of constraints that ta over behavior and this I've demonstrated by talking about Med Nativity and the like um you have uh how to say you have added Nuance in the uh notion of constraints which is that a sorry again AVL from your side is just I'm not hearing any audio ma or Maria do you hear him no nothing you can't write over Collective Behavior so there is scale of one specifications yes all right could you repeat could you repeat the last 45 seconds

can you hear me now yeah okay so what I'm saying is that the engagement with cultural landscape parentheses it is dual to the existence of shared constraints which act over individual Behavior because when I engage individually with the landscape features that is a characteristic of my local material and cultural environments uh the likelihood will drive my behavior my intention my presentation in a specific way and instead of looking at how each individual agent engage with the landscape I can look at the constraints that emerge from this uh Collective activity so is that do you hear me now okay yeah yeah okay good and uh so let us get back to the uh question of what makes us human uh we had talked about human specific uh mechanism phenomenon that seem to be involved in culture those were language and Role taking and those two phenomena seem to be pretty robust um candidate to explain the coherence of a shared cultural landscape or set of social constraints uh language it provides a space of Public Presentation like representation in the classical aesthetic sense that you can negotiate that are actually out there in the literal sense they are sensory signals that are in the environment and that uh structure agent Behavior so when you talk to yourself it will constrain what you pay attention to implicitly what you can and cannot do and when you talk with others this will influence what you say and what you want to say and what you perceive to be right to say and so it will constrain your behavior and in a much stronger and more robust way than you could have to implicit intentionality inference and Role taking in a more subtle way uh brings agent to think existentially to like not ask the question what I am supposed to do in the situation but what an agent that is embedded in the specific cultural context where I am embedded is expected to do and so this basically forces you to um have a synchronization between what you experience as possibilities for action and what you expect our cultural expectation over Behavior so these two mechanisms there would be very very strong uh mechanisms to enforce the coherence of cognitive landscape between agents and it just happens that there are things that exist only in the one species that has full-fledged cumulative open-ended cultural Evolution so yes I think this is uh interesting okay a you just unshared your screen sorry you unshared your screen or is that done uh no I did not un my screen hm could you just like reshare it yes there you go all right all right looks great continue thank you nice um so yes the question is um as I said there is a duality between each agent engaging with a landscape for action and uh constraints that exist over the landscape that is shared that is straightforward but uh in the set of constraints that is um characteristic of a given landscape uh basically to account for cultural Evolution this must be a set of constraints that uh changes through time that builds new constraints and new Deal flow and it's

something that you cannot get with a landscape in the uh sense of deal System Theory that is not Nial System Theory does not deal with systems that do aand Devol so the question is what relation can we make between the Dynamics of enculturation according to each agent integrate specific cultural expectation and the Dynamics of open ended cultur Evolution through which cultural landscape and the constraints um like constructs itself through time and this is where we get to the more holistic side where uh I think we can see why the specific construct of constraints is a critical as compared to the simple cultureal landscape approach let us take a look at this there is a thing CA called biological autonomy it is a quite recent term in its formalization the relevant formalization that describes the ability of a living system for self-construction like maintaining one structor and changing it through Developmental and evolutionary time and the best formalization we have that so far is from Mon and and it's a constraint closure a con constraint is simply a relatively slow physical process that shapes a faster process like canaly material or information or energy flow without itself being affected dur at the time scale of the interaction and for mosio you get biological autonomy where you get a set of constraints that collectively produce itself each constraint constraints is produced by other constraints in the set and each constraint produces other constraints in the set this is a very simple ID this is a very informal ID in a sense that it is not doesn't have Dynamics doesn't have a physics it is just a formal statement of what we mean by autonomy and this uh I basically is the best grounding we have to think about what is living and what is an agent although in itself it does not entail agency and uh when we look at Social systems or systems as a set of constraints um what you can see is that system such as the brain multiscale organized system they can be read as basically a set of n constraints like your uh dmn the activity of your um default mode Network which is the basal activity of the brain in the absence of stimulation it's uh basically is consed of dial constraints over what the composant um tissues do at a given point in time and this itself is a constraint over what individual cells do uh through time which itself is a constraint over the evolution law of an individual cell etc etc etc so I can always look at a m scale system as a set of nested constraints at Ned scales and in systems that are um autonomous that are biological cognitive agent um not only the um smaller scale uh systems like cell they can uh they enable they construct and they enact the constraints that are AOS scale agents but ioscale agents can also shape back the behavior of low scale agent like of course if I have you know if I do something which is something that I as an agent do this will constrain my the activity of my neuron networks this is like heaven learning this is common to all model of learning learning goes through change in network

structure and so there is an opportunity to model uh emerging agent from like Collective network activity and to look to look at interaction between description at Ned scales of behavior and uh this is applicable very fly to the human species so this is a suy States and something that you can look look is that uh it is very dense it has very dense building that are buil in a specific way it is near a river uh which enables the um trade at Great distance it has a bridge so entry to the city is constrained to a specific point and it is essentially surrounded by several Fields very flat lands that are Rubble that are um fertilized probably by uh water from the river and basically all of these feature are enable the production of a state societies societies with a central Administration that does stuff and these societies will mostly produce context uh material context that faize their own Survival for example then settlements enable you to like have everyone together to tell them what to do temples which give a matal grounding to the authority of the states uh bridges that allow you to move things around and to control what gets in and out and more cultural seral fields that allow you to like take the food and bring it back to the city and the prod ction of such settlements and such material environment by States is uh somewhat robust Intercultural Fon uh at more like ancient are a multicultural scale um you have a set of process that are critical in the construction and action of norms in humans and most likely common Anor to to those processes is dancing and singing dancing and singing that they give you um like embodied synchronization and this is shown to drive development of group identity and this gives you basically emotional grounding of the Norms that are shared within Society because they those Norms they are not abstract expectation they are uh embedded in your specific emotions in your specific embodiment while you do things that are uh critical to your ecology and ritual activity and storytelling they allow uh more complex storativity by using you know symbolism and the system of toity that are maybe less embedded embodied sorry so less embedded in the uh like On's direct experience but they allow a more more complexity in the nature of the Norms that are transmitted and enforced and uh the these Norms the by the simple fact that they are perceptible through matal symbolism and through storytelling they become a causal force in the cultural landscape in the that is shared by a group member so they become thing that are uh the power to produce or constraint the framing in which people will read specific situation and therefore constrain and produce s behavior and this um this is like I've said this already so this looks like repetition but I want to hammer this because if this is the case uh then we can be interested in what complex uh tals in so sorry in agency that emerges from the constraints embedded in complex Tales so we know that uh people can perceive nonmaterial agents under

certain conditions uh this is common in schizophrenia like you will hear voices that tells you to do things and that are not embedded in a specific body this come on in a deive Valen disorder because you will have agents that basically share your body while while having an existing while not being tied to a specific uh body like one sorry I'm misspeaking one specific personality can exist while they're not in control of your body and so in a way that is is not contingent on the direct matal inaction of what they want and want not to do and under a lot of anthrop anthropologically um described conditions uh that are not considered pathological you can basically have regimes of attention that produce a normious agent that you can talk with and that have motivation and that have traits which are often called tps uh which is term borrowed from Buddhist literature and uh so this emergence of the material agent how to say if I consider an agent to be something a closed set of constraints and I consider things like tail and ritual as mechanism that produce constraints in Behavior there is basically no difficulty whatsoever at concept level there is is no unexpectedness or no complexity to say that uh specific systems of constraint that happen to be closed and that are embeded in those tales and rituals they exist as biological agent and uh so we can have basically a model of gods of nonmaterial Agents which is a better less loaded way to put it that relies on a systematic training of your regime of attention and of course uh when it's Collective it will like those agents will be social agent there will be things that have uh Power that have the ability to frame the debate on specific things if you and I think that uh so this is sorry the painting that is on the right of the screen is a painting by William Blake that presents uh basically a late 18th I think nation of God which is um not really old testament which is a bit more New Testament but which is mostly in sensation of the um clock maker God that was popular at the time and what you see uh on the painting isan So William Blake's version of God that is in the sky like very powerful very Z like imagery and that is using um like this his hand to draw a compass that is considered a symbol of like um constructing things of masonry like the compass is one of the Freemason uh symbol for the same reason it is a symbol of orderly Creation in Willam Blake's uh paintings and so if we share a belief that there is a God somewhere and he build the world as it is intentionally and this God is benevolent or at least much too powerful for us to fight with basically your option they B on to following the order because if you don't God will remove you from the order and you don't want that do you so even if you don't have uh the existence of like um matal a materal sensation of this God even if people don't really really believe that there is actually a guy with a bird that does stuff uh they sh ritualized uh belief in an entity that has this specific motivation and the specific traits it will uh shape heavily the

social landscape that is available to everyone and this is compatible with the idea that it does not entail the existence of actual agency that is tied to this specific entity through the constraints that this entity concretely is and something that I find again interesting is that religion that are popular today so Christianity Islam Buddhism Hinduism if you think of something as a religion will likely fall under this this um handle are typically what White House called doctrinal religion which are religion that are where basically you sit down and you look at rituals and there is uh priesthood that is trained to do the ritual in a specific way that is supposed to mean specific things and this is basically optimal to build very robust notions of uh what gods are what they want from you what they do what they are motivated by etc so I must insist that this is with a question mark But in principle the um the social constraints model it can produce an theistic theory of gods and nonmaterial agent sorry as socially embedded nonmaterial agent that are computed uh collectively through engagement with a specific cultural landscape where uh ritual and doctrinal um elaboration are important components and I can say with exclamation mark that this model it enables naturalistic theory of social agency that is not tied specifically to individual bodies if I consider a social system actual organization to be autonomous that is closed system of constraints uh then agency or maybe um non-trivial agency it simply reduces to the power to shape sorry shape this uh landscape and you can have uh emergent social things do that you can have states which follow specific rules so social constraints and specific Norms so G constraints that have autonomy and that are able to shape the landscape but in all of this like this part in my opinion uh shows that c constraints specifically are relevant to study the social worlds that they correspond [Music] to relevant social process in human other agents that are institutions and maybe again question mark material agents sorry agent institution so Gods uh the question that needs to be asked is how we can model the portion of constraints because all of these rely on the IDE that somehow in the landscape that we experience there is such a thing as constraints and they are well defined thing that exist over Collective behavior and that they have the property to construct itself and for reason we'll very shortly see to frame this mathematically in a coherent way is insanely hard I don't know how to do it no one knows how to do it we have leads but they are leads nothing else and so we'll turn to the question of uh whether active inference frames a gge theory for inculturation ination you should know what this is by now go Theory we will see that soon so to explain gge Theory I need to explain explanation which is fun but mostly symmetries uh in physics in physics uh we frame physical problem around symmetries symmetry it is an algebraic so something that has composition like plus groups of

transformation over the space of possibilities which are irrelevant from the perspective of the same Dynamics is very strict so let's look at symmetries this is the ceiling of the Lola mos in found what you can see is a symmetry under the C_{32} symmetry group if you shift this picture by 2π one over 32 uh full turn it does not change this is what we mean by a symmetry this is a transformation a set of transformation that is closed and that does not change well not a picture of course in the context of physics but uh the Dynamics of the relevant system we are studying and for formal reasons I will not absolutely not get into here explanation in physics it usually derives from the statement of symmetry for example when I state that energy is conserved this is the same formally stating that my system is symmetric in time when I state that momentum is conserved that radius formally uh to stating that my system is a symmetric in space okay so most things in physics can be framed as simply symmetries over System Dynamics and a gge theory is simply uh elaborate instance of this core ID so if your symmetry is uh has two properties first it is continuous so not like the uh a set of small rotation but like the set of all possible rotation that you can take uh around the circle and local so you can apply the Symmetry transformation to every single point in space uh your problem basically can be framed with the real degree of Freedom so because of the symmetry does not change a system whether your system where your system is in the for example Clockwork rotation it is not important it does not change the dynamics uh so this R degree of Freedom we call a gge parameter um the core feature of gge theory is to take the geometry of that problem and make it into a physical Force so um when I say uh G parameter so this is um you should see in the in the screen uh non-trivial Z of a two Z principal bundle of a circle which means basically that you have a circle it has two stages and when you are over it you can be either on top or down and you can't know where it is so this is a not quite to the gra freedom because you have only two choices to get to the gra Freedom you have to get a continuous space of choices but this is basically what we mean by gge parameter it is a geometric property of your system that you know mathematically exist but that cannot possibly be observed or how to say corresponds to a choice in how you absorb your system so this is why we talk of G because this is a choice in how we end the system but I will not get into the details of this here and the thing is that because of its redundancy The Gorge parameter does not have to be the same in a specific trajectory uh at any given point in space and from that you get curvature and curvature if you are anywhere in physics and you know a okay last thing we heard was if you're anywhere in physics know about relativity Theory you will know that to the like naive embedding space so if you know uh vaguely Theory you know yes sorry just if you're anywhere physics and

then continue from there yes so if you are Vaguely Familiar with relativity Theory you know that curvature in a space it corresponds to forces so in relativity Theory the curvature of SpaceTime it corresponds to uh the gravity force and generally in uh physics the curvature in other weirder gge spaces it translates in the other FAL for so weak Force strong force and electromagnetism so yes just stating that there is a space which has a Gorge parameter in it and this space has a curvature it entails that you have a force that emerges over the let us say natural framing of the problem and uh this uh speaks very closely to activ inference so active inference can be framed as a consequence of a formal result that is called the principle let us uh write a dial system stochastic which means that it has noise in it and which can be a partition between an agent a environment b and a boundary sorry an environment E and A boundary B such that the distribution of of the agent and the adment are independent from each other uh given the disposition of the boundary B which kind of means that any information uh in E that is relevant in a and vice versa has to be in B but doesn't quite mean that but it's very close uh then you have a information geometry which is uh like geometrical construct where each point in the agent corresponds to dision of belief over uh the space of external States and because of this information geometry each possible internal state of the agent it entails implicitly a belief over external States so this means that given a condition of partition ability uh I can write basian belief over my environment as a simple geometrical property of the space we are uh working in and as I just said when you have a geometry and it's curved you forces and so uh the F can straightforwardly be framed as a go Theory where the Symmetry is in sensory entropy technically but let us say the space of belief I am evolving in directly constrain my action in a specific manner to constrain basian coherence between what I expect to be the case and what I experience to be the case and so straightforwardly this grounds active inference because like what I just said is what I said to explain active inference up to some technical words that were added in but yes this result entails uh predictive processing and predictive control of behavior of the agent and to some extent impr proves the possibility of this existing in bare uh demicol systems Theory but I do not think that the FP can possibly represent enculturation so when I talk of AAL system this is a system with Dynamics in it you are I think you understand that I do not think you understand the formalism that we imply by ital system is a flow function F_e a Time parameter T and State Space X so what that means is that to frame an equal system I have to write a space let us say I don't know just the Olan space with three dimension I have to specify time which is like naturally in the CLE space what we think of as time and I have to specify a specific way in which the um my

system will move in space in a given uh step of time so fig gives from any position and any time slot where the system will be if it starts from this position after the specific time duration this like this is what AAL system is but if you have noticed this uh this entails that I Define preemptively all of the space of possibilities of my system if I have a system that is a you know open-ended system of constraints that can build new stuff that can build language that can build accounting whatever new States this doesn't go maybe you say uh We Could reconstruct a minimal motion of new stuff Happening by work sorry I'll ask him again a again repeat the last 30 seconds but your audio is still not there it's like a 30 to 40 second period where you're audio just doesn't transmit none of us can hear it but the internet is fineing over the dyamics of the system per in system the so you can do that EI so yeah continue yep continue okay so uh let me system can be written as a flow function F_e over a state Space X and A Time parameter t uh when you specify X you specify the total space of possibility of the system so if you have a system that has openend evolution in it well you cannot work with a fixed space you could maybe uh like circum the issue by implicitly having new things happen from a change in the Dynamics of the system but in theal System Theory the dyamics are pregiven through the flow function feal system system AAL System Theory system is a specific deterministic flow so you can't change the flow through time so you can't model the construction of new constraints and you can't model the emergence of new parameters that affect caal your system so the FP in of itself cannot present the creative evolution of either cognitive kinds so such as money shares or soci constraints and this is where we have to get in weirder uh spaces formally to get something that looks like could model openend Evolution although I cannot say definitively it can because we didn't yet by construction when we work with set theory which is the basic theory of mathematics which is about specific objects that are purified we can't have what kofman calls adjustment possible which correspond to new evolutionary possibilities that are brought about by the evolution evolution of evolutionary system like living systems um simply set theory forces you to Define all of the possibilities at step one of the modeling process so can't have your system that builds new possibilities with activity this is not a coherent Concept in set theory however what we can do is to look at rather than set theory at a category Theory which is let us say another approach to mathematic which looks not at specific objects and their relationship through functions but have specific spaces and the relationships through functors and this change in approach basically does not entail at least not a Priory that when you write a system you have to write all of the possibilities that are within the system preemptively you can write to space and then look at

the relationship uh between the space and other adjacent spaces that could correspond to adjacent spaces and one thing that is related to that at least in spirits is uh I won't get into detail but formal approaches that allow you to build Quantum State spaces from the property of quantum objects rather than the opposite so simply speaking we can look at the problem of uh how cognition brings about new possibilities not as a space of possibilities in which there is a flow but as a set of spaces in which specific events uh such as observation such as enculturation such as invention of New Concepts can bring about new spaces and when we rewrite activity difference in the language of category Theory we get things that are uh that do not quite say the same thing as dynamical System Theory like actual active inference actual Feynman principle uh but we get things that are very very complicated so the diagram that you see on the bottom left of the screen is a commutative diagram I could tell you what the commutative diagram is but I don't know so I could not explain that is meant to present the alignment in reference frame or the mutual active inference between two subsystems so again formally we have no reason to believe this can't model and evolution but also given the level of abstruseness of the math we have very very few grip with which to understand what we're looking at and how this relate to specific cognitive and cultural processes so there is a lot of progress to be made in that direction before we have a self standing theory of open ended Evolution uh and uh the basic handle of constraints I think is a good bridge between the very concrete very simple dynamical System Theory and the Very abstract um problem of M Evolution so let us wrap this up here and uh try to take out a few snippets of actionable information that was included in the talk uh first agents by Nature they engage with the world with their world as a cognitive landscape that is defined by the modalities by which they perceive the world that is defined by the intrinsicity that the agent experience and is again sorry just back 22 seconds but still the audio is not ready yet but it's okay we'll breathe before we jump into the conclusion okay I still don't hear no we can do the slide game where someone else presents the slide any volunteers okay yeah I heard that okay cool can please continue okay we'll try to go fast because this is not agreeable uh so one agents engage with the world as a cognitive landscape which is defined by our forenses so minimal presentation of the action that are possible in the landscape and the normativity that's embedded in the possibility of the action two uh the fact that humans collectively enact relatively coherent structured cultural landscape it entails that we have social constraints over Collective Behavior and the existence of those social constraints they enable AI thought do not necessitate the emergence of higher scale agencies such as institution organization and maybe Godlike beings and uh last the dynamic of constraints they entail a

duality between the creation of meaning so the way we human uh create a way to understand the world and the construction of physical possibilities posses in the sense specific concrete causally relevant possibilities uh that characterize a specific social cultural organizations and maybe like physical obervation pareses but just not worry about it and all of it is embedded in this diagram the engagement and the uality with a system of constraints that produce itself in a manner so a claim that I'm making on the record here is that if we understand correctly every arrow in this diagram we understand cure Evolution we understand what human are and we understand evolution in the cultural world and here I'll go through the slides in case someone needs a reference and has the time to you know go through the video to find the reference so thank you and sorry about the audio thing did did this go well yes it gave some much needed silences so it was all good um cool well first ma or Maria if you would like to give any comments or or Reflections otherwise we can go to a few topics and then talk about how we want to conclude with the course but first Ma Maria do you want to say or ask anything um no I mean this was a great summary uh I think you covered a lot of grounds I think uh ideally I would have liked to see um how much we can do with Gauge Theory and how this can uh potentially link us to uh and understandings of power I think that would be uh that would be interesting cool thank you Maria all right see you Maria okay anyone in live chat can ask a question or um let's first address the the course question while people are writing questions live chat so what what do you see for the coming weeks as we kind of conclude this first round uh so um you're asking me right so uh for a reason I do not understand and Danielle do understands uh no one came up at the M discussion session so I thought it would be interesting to uh like push the for the last session to have you know everyone saying their things have feedback and uh I have a current thing uh to move on from uh the question is whether you need you feel there is a need for a discussion that is specific for the session we have we have had just now because lots of thing were covered lots of audio problems were existing and basically the two options are either next session is everyone present uh we give you feedback ch or we have a discussion session so at the time that was initially planned for the conclusion and then we have yet another session that will be the conclusion where you will make the presentation and you will get the feedback so we'd like to know which is your preference because all both are very possible in my book like doesn't change anything to me so what you want is goes yeah now also open to your thoughts and of course we're sharing it so that people can write in live chat or let us know in the coming days what they prefer I'm down for a discussion of what we covered here the slides being linked and going into some of these next

steps and then a form based registration for a presentation only session so that we'll then know exactly who wants to present what and we'll then we can ask how many minutes they want to do or something like that but ma uh yeah I think uh it would be good to get people who are interested in getting more active to um to come and make a small presentation uh explain what they understood uh describe the highlights for them and potentially link it to some Topic in social sciences we may not have covered here uh but that they now see through a different lens I think that would be interesting okay all right I will uh look through the live chat while I was getting some water and then we'll ask any questions that people come up with is there ever symmetric information between or among social entities or maybe from the scripts perspective ma or from the or however it's you see it in relation with the constraints I mean what about asymmetric constraints I'm not sure understand the question can the person give a little more detail in terms in the question okay um's back but uh there was a question in the chat about um are in can information ever be symmetric you talked about symmetries so are those symmetries in the information that two social entities have or are they in the constraints that entities experience but what are the symmetries and what are the breaks I would say that depend in which space you model the problem an information it entails uh symmetry because it has to be readable so you have to be a conservation between uh context or time of something that is considered to be the content of the information bearing thing uh but also it is a symmetry breaking because uh if you take any like uh presentation format let us say binary a random string means nothing and you have to break the symmetry of the UN distribution to say something so like it depends very heavily on how specifically you are modeling the problem but like for fundamental reasons any uh mathematical representation it depends on the implicit model of the symmetries of the system because you want to things that do not matter but you have to things that do matter and then you have a very uh uncomfortable amount of Freedom over how you want to present any single thing certainly this comes up in learning active inference and in talking about it like what you described how many ways there are to model a given thing which is related to which symmetries are preserved as you mentioned so H how do we reckon with that space and with the different ways of representing the same social setting so um a reason why I'm not quite comfortable with the uh notion of information is that in all possible settings where I Define quantum information Theory as not possible because like very few people overall think about it information will be a probability distribution it will be a like a functional of a probability suion that is minus to the logarithm of the probability that is what self information is and this is defined over probability distribution

distribution is defined over State space so implicitly information entails definition of a state space so if you want to work over things that are prior to State spaces you cannot work with information in like any coherent definition so the basically the one thing uh you could do uh that is vaguely inspired from work in a Quantum information Theory mainly by Christopher first is to not look at the probability but at the um sorry not to look at the information I'm using that term in a cial way now because I have to switch meaning uh defination not as a property of the distribution but a property of quantum operator that corresponds to observation and that embeds basically we could say again in a very very colloquial sense something akin to an expectation or a measurement uh or a synchronization event in the active inference model and this for a specific type of operator uh for a specific type of spaces you can get basically the space that flows from the operator rather than the other way around and then information is like it's not kov complexity it's not shannel information it's not a variant of entropy it is basically what hypothesized by the specific operators that you use to interact with matter and with cultur and natural environments and maybe maybe there is a big maybe here I should say maybe one more time maybe this Con uped in some way uh to account for the process of eneration where the um basically the process ination is considered as the integration of specific reference frames like expectation cially and this entail the evolution of a cognitive landscape and our social constraints either defined over the cognitive landscape or defined as a self-standing mathematical object I do not know uh but if we have that and if we have a decent level of understanding of what constraints the evolution of neur frames we have a self sending theory of the unfolding of ctive Landscapes and of openend culture Evolution this is a sufficient condition I am sure of that but it is very very hard to reach I there are so many threats by which you could take it and each one is a like the level of post do in mathematics and I'm not talking about 19th century mathematics not talking about algebraic topology I'm talking about weird category thing that people only figured out how to define 30 years ago so yeah I'm not confident I will give you an answer to that within five years I'm not confident wow very interesting there all right some questions from the chat so first upcycle Club wrote do you think ritualized storytelling is driven by social constraints everything is uh driven by cir constraints we just Define cir constraints as the Contra that exist over behavior um if we expect that every night we have to go around the fire uh and listen to someone tell a story then yeah like this will happen if we expect that we have to go on YouTube and listen to talks about abstract things such as inference this will happen so in this sense everything every single activity you have is embedded in a landscape of expectation and this is why it's pretty hard to

talk about basic mechanism of transmission because which whichever cognitive mechanism you are looking at is encultured and could in principle and likely has been a few times uh challenge Evolution so um yes cool and that takes me back to the scripts discussion in papers the strong script at exactly this time here's the script that we're all going to read from on through the spaces of weaker script like we tend to meet here or there happens to be a path here okay so then some comments from M Lenin I thank you I don't know if making a presentation to explain what one understood is an attractive invitation still two places I did not track as well as I hope was one dissolution of agency as we know it and two the informational geometry among agent environment and Boundary went over my head thank you so dissolution of agency and um agent environment boundary yes I will get back to the slides um are you seeing the window yeah I'll I'll I'll bring it in that's fine thank you okay no no so uh the dissolution of agency is um basically the idea that when we when we think about agents people like we think as a system that as a body and that has a brain and that has the ability for discursive like deliberating over what you do and not do and we place agency at the level of discursive beliefs so if I say um I want to grab the water and I grab the water I grab the water out of my agency if I inadvertently like I don't know bump into something and make it fall or grab without thinking about it is it agential uh not most people would say yes but that's actually not in line by the most used model of agency like legally speaking you are held as uh not responsible or less or less responsible of things that you did not have the opportunity to debate over but from a purely cognitive perspective and if the inference model is correct the uh bulk of the intentionality of your ability to direct your own flow that you have does not come from reasoning and deliberation and uh Central executive system like cognitive function that we associate with reasoning and agency it is it comes from the basic expectation you have yourself and the world because your expectation over the world uh constraint heavily implicate the opportunities for Action that you perceive and your beliefs or yourself constrain just as heavily um the options that you will end up pursuing and so the agency your ability to decide of stuff it mostly reduces to practice process over your self model and your Role model and this is while not like I I don't see any problem formally or empirically with that picture like as I understand it accounts very well for for the for the um neuro psychiatric and like psychological literature and sociological literature like this is good this works scientifically but this is counter intuitive because again most people would expect you to have reasoning and then to decide of stuff and then to execute of all the stuff you decide which again I don't think this happens so harder question is the information geometry I kind of gloss over it because yeah so

uh the information geometry is a geometry that is defined within AAL system equipped with the partition blah blah blah and so by a geometry uh I mean basically a space uh of curvature that exist over your bzal space this is basically the same thing that I'm presenting in the um uh bottom left of the screen here you have a space that is your normal space where you see stuff happen which is represented here by the line and you have a curvature that is defined by some of the things that are defined over the space and by the freny principle uh when you have this partition condition that is verified then uh each space uh it's sorry each state in the agent internal uh States like each specific point it can be corresponds to a belief distribution over the um states of the environments and so this is in this sense that we have an information geometry like we have a geometry so curator over space where the space is the internal state of the agent and the geometry is belief over exal States and uh furthermore this matters because uh the agent like is actually constrained by the the agent is uh effectively constrained by the by the geometry in such a way that it inter its internal flow uh optimizes the Bas and coherence of the belief that exists according to geometry So This is highly this is math this is like straightforward but lots of words lots of rection no real visual AG put it so this is more something to like read about and something to hear about which is why I gloss over it but I hope going over it again brings a bit more of clarity how would you connect what you just said to the differences with written and oral culture like what is hearable what is seeable How does writing in the printed or etched or digitized form relate to how you discussed culture today uh I I see you as Frozen Danielle can you hear me I can hear you Ma okay so do answer or do ma answer you want yeah please I can give a little bit of an idea but effectively we're talking about precision and temporality right so writing and uh discourse the discourse has to relate to pragmatics and the the discourse is likely to be interpreted because it has lower Precision than say writing has over time um so here you're talking about the ways in which the information geometry will be bent and shifted through the Precision of the embedded um of the embedded normativity that is given rise to uh through these um through these markers that you can point to over and over and over with high Precision I think that's the link yeah basically that uh so when I talk about Ro taking and language I claim that the core reason why this matter is because this enforce coherence between the cognitive landscape experienced by agents and writing changes massively the scale I which it is possible and it changes massively the level of emotional embedment of the uh of the of language it is much lower in speech than sorry much higher in speech uh than it is in uh writing especially if you account for like singing as a form of speech uh so historically writing very very clearly and very Ambiguously powered the rise

of administration so States so organization that uh do a lot of accounting and bookkeeping like book is literally in the word and that enable the centralization of resource management uh this is explained in detail in uh the logic of writing by goody g o o d y and yeah basically this writing is a core cultural Innovation that enabl centralized policies city states things that yeah that's very interesting so one topic out all that we've discussed and maybe you could unpack a little bit here is thinking of culture and these cognitive ecosystems leads to consideration of the cognitive security or of different information risk aspects so where does that come into play with the formalism or what you discussed here so uh from a cognitive security aspect one thing you can uh read in formalism quite directly is that if you control the environment as in the cognitive landscape experienced by agent you control what the agent do uh directly so there two things that are very very very um stereotypically ass cognitive security this is uh Russian uh propaganda not quite propaganda cognitive Warfare campaigns pretty much around the world uh more generally the um propaganda that is used by authoritarian regimes to get stuff going and the um uh Cults yeah cults that uh historically are uh tied to the worry of brain control by evil CS that can control you from a far like you had a you know telecommand and so in both cases um actually let us get back a bit when you are a state that does propaganda uh and like when you a state you can do a lot of materal engineering so producing the Environmental uh states of the um in which people live in and uh for example if you look at the city of Brazilia which is uh one of the uh very few uh spe specifically built capital city that exist in the world there is like one in there's Brazilia and there is like I think maybe four others and those cities are basically very big glorified administrative centers and brazilan in particular it does um neighborhoods for work neighborhoods for ler neighborhoods for um for habitation it is extremely uh divided and because of this dividedness you basically don't get to do stuff that are not work or hang out with your family that is not a possibility and of course this Chang massively the way people experience their life and this trans massively the topology of thisal network that can emerge uh but when you are doing cognitive Warfare like in the sense that NATO defines so you are state or state lonization and you are trying to with another state or state oranization you don't get to just build cities that are bad but is not possibility that you have so what the um Russian do in a systematic way is to try to basically create doubts over what information is reliable so that people feel paralyzed uh the target I must insist critically is not the information that is available it is TR that people can have the actionability that the people can give to the information and so I tend to propagate a lot of conspiracy theory nonsense because it occupies a lot of space and it prevents the possibility

for public debates to occur and another strategy they use is to manipulate the personal identity uh that are embedded often in ethnicity or uh in the case of the US of advanced countries that have uh where political life is mostly decoupled for for ethnicity and I understand the irony to say that this is the case in the US but you don't have like ethnic ethnically specific parties are armed forces so this is less the case that elsewhere so in these countries they will use political identity so they will use Republican identity uh or leftist identity but mostly replicant identity to like just push complete nonsense and get people to act in a way that overall dis stabilize the grip of the us over uh their own population and so this is the basic uh way that we do cognitive Warfare and you can understand this in terms of social constraints because again the target is not like information of itself it's the it's the possibility that is afforded by the social context in which you evolve at turn in cent the goal is to create uh doubt and noise in the public setting is strate constraints that are not compatible with a efficient Collective organization and a way like a way where you do it is um when you see it being done in a much more systematic and efficient way is of course uh with intrastate influence where State administrator and state institutions they try to create grip over a population and if you look at the birth of nationalism uh like we have the notion today that nations are somewhat coherent political group that let us say the French hang out together and take the ion together uh this notion it's not quite recent streetly speaking but the fact that is dominant that it is just a St expectation is very recent and it's it derives from the French uh that somehow where a group of administrator somehow managed to convince everyone that they shared uh identity and they shared Mutual obligation responsibility in virtue of this identity and uh the way in which this happens is they sent uh instituts in villages and they sent soldiers and you had to listen the the children had to listen to the inors explained to them that they were French and they needed to do French things like going to war for friends and if you did not let the children uh like do that it you went to prison and no one wants to go to prison so with of course I am a I'm doing a very very huge caricature here this like there is not a shadowy set of influence sense this is a over specification but what happen is that the fren State managed to create a social Niche up in a very very forign bra Villages where um French Hood became a basic condition of existence most of your social thing the fact that you were legally recognized as existing uh the activities activity that you had lot of those were embedded in your perception of being French and so when this is the case and when you see that Administration is tends to be on top of things thetion tells you you have to go to war in virtue of freshness then you go to war don't like fight against the administration you fight against whoever Mission tells you uh fact that you have

agreed implicitly that you are French before hands makes this lot lot harder to resist basically than if like let's say just uh send solar at you and kidnapped you and give you a swear until you fight this this is not how you make things efficient uh Army this way are unorganized they are unmotivated they are not good and if you want people to have stakes uh you need to give them power or you need to give them the impression they have power because they are French and here French uh like govern things and whether you are in the set of French that actually direct things or not bad question to ask just you know go to the trenches and this type of uh things we would not think as cognitive Warfare because we would not think as a like it as a between like implicitly when we talk about Warfare we talk between states or between Nation conflict and this is within a nation within AAL group but you can look at um Cults and how they create social influence uh you can look at for example one of the most famous Cults that existed which is the monies or the unification Church um they were widely accused to make uh brainwashing they were the accused to somehow control people and make them do things they don't want to do that is not possibility you are action like you cannot kidnap people and hope that it will be okay you will get killed by police if you do that most likely in the US in this specific um uh space where the ination church evolved what you can do is to create a context where you have a huge grip over what you what people can and cannot say what they can and can think about themselves you can tell them that uh Your Leader is the uh coming of Jesus that uh is going to bring everyone in paradise and you can make people go to a work session to talk about it and you can just like physically cling to those people all the time until you are quite confident that they feel that whatever you are doing is important and you can have very big emotional ritual where everyone like uh cries and become extremely embedded in the uh group ership and who can be extremely supportive of group members uh while being extremely like aggressive to anything that go away from group identity and when you Pro this environment and people Eng in this environment for a long time right time to develop this group identity so when you tell them uh the big leader tells you to give us 2,000 Euros uh sorry 20,000 20,000 Euros for your ancestors not to go to uh hell that seems reasonable that seems like a reasonable ask which it would not unless you had a very tight control over uh the social environment of someone because then it's not just giving money uh in exchange for good this is about upholding your self identity as someone who follows and believes the C stitching so and yet again this is something that is formalized directly in terms of s constraints because this is about what you can and cannot do in a given text this is not about like abstract belief it's not about brain control it's not about anything but what you can cannot do in a given

social context all right there was a ton there let me let me kind of retrace a few of the pieces that that I heard there so towards framing cognitive security under this cognitive constraint framework that you're discussing for which active inference under the free energy principle is the process theory that gets it done first you described how modern states have a lot of administrative capacity and affordances especially through writing and especially again in the last decades and centuries of information environment changes and then you pointed to two kind of patterns or genres or motifs of influence or cognitive Warfare propaganda and Cults discussed one example of them surely could have gone into more detail or there's other examples and perspectives and so on and um as ma alluded to earlier Precision modulation is a key aspect of constraint I mean Precision modulation of how tightly you're holding the baseball bat is Rel the movement its Precision in wobble is going to be related to the to the grasp and so um increasing uncertainty can lead to anxiety and doubt all of that's very wrapped up in a narrative setting then you discussed how there's a ratcheting of thinking and acting like a state breaking symmetries like states do and getting grasp over the subject over the citizen that's a complex topic but that's what you went and then there are asymmetries around control of the environment the information environment ability to to read write edit in the information environment and that leads to these uh engineered distributions of what is important or what means what which justifies or even writes the blank check for requests of funds whatever the narrative is around those funds um but more fundamentally those strategies can threaten sense making which is to say the inbound perceptual component of AC of inference and then also decision making and action selection which is the fundamental complement the outbound aspect of active inference I'll read another comment from live oh yeah do you want to add anything else on that get uh like I'd like to point out a very very specific example contemporary uh which is the um uh like the campaign by Israel to remove people from the Gaza bands uh so just I don't know when people will watch it so on the 7th of October 11 days ago Hamas which is a terrorist organization islamist organization that is dominant in the Gaza band attacked Israeli and slaughtered IR settlers that were nearby and slaughtered tourists that were nearby and attacked military bases they made I think 2,000 deaths and uh so the state of Israel has a cont that is currently um led by very far right parties things that we would not have equivalent in of in Europe or the US has decided to answer by uh like blaming each individual Palestinian for this and trying to uh like Deport them elsewhere and uh so like this this is not itself what I'm trying to comment about what I'm trying to comment about every single comment you will see about it on Twitter makes the confusion

between Hamas and pales on one side and IDF nanu and the israelian on the other side and so this produce of framing that says basically um people are not people are trying to decide whether the amas action are jifi or not and implicitly in this debate they create a context where whoever lose uh whichever uh group lose the Legacy debate gets to be killed to the last and so you that produce very very um basically that produce a framing that uh makes the only possibility in the conflict examination because each side uh like there is no like this is not two side dyamic there is not Palestinian Land There is the uh the Israel land there is the um I will not make all the relation but by this framing uh like the framing that either the Arabs or the Jew must Prevail and kill the other you give a lot of power for groups to just make this a genocidal war and justify things like cutting water to one million people because uh like 200 of them uh committed massacre and like I do not know if there is a single actor that tried to get this I do not think this is the case but this is a very powerful instance of how proa does not need to Target what you believe but it does need to Target how you frame problems yeah lot to say on that any last pieces you want to discuss uh no it is good I have talked a lot I'd like to stop talking okay well thank you we will um send out updates to people about the coming weeks so thank you all right bye ciao thank for thanks for your attention goodbye